

Artificial Intelligence using Python

Hands-on Exercise No. 2 DigiSkills 3.0 Batch-03

By Bakht Zamin

Data_Preprocessing_with_Pandas

Problem Statement

Task 1:

Dataset Loading:

- Load the dataset into a Pandas DataFrame
- Display the first 10 rows

Solution:

```
[ ] # 1-Dataset Loading

# I. Load the dataset into a Pandas DataFrame
import pandas as pd
df = pd.read_csv("HousePricePrediction.csv")

# II. • Display the first 10 rows
print("First 10 rows:")
print(df.head(10))
```

First 10 rows:

	Id	MSSubClass	MSZoning	LotArea	LotConfig	BldgType	OverallCond	\
0	0	60	RL	8450	Inside	1Fam	5	
1	1	20	RL	9600	FR2	1Fam	8	
2	2	60	RL	11250	Inside	1Fam	5	
3	3	70	RL	9550	Corner	1Fam	5	
4	4	60	RL	14260	FR2	1Fam	5	
5	5	50	RL	14115	Inside	1Fam	5	
6	6	20	RL	10084	Inside	1Fam	5	
7	7	60	RL	10382	Corner	1Fam	6	
8	8	50	RM	6120	Inside	1Fam	5	
9	9	190	RL	7420	Corner	2fmCon	6	

	YearBuilt	YearRemodAdd	Exterior1st	BsmtFinSF2	TotalBsmtSF	SalePrice
0	2003	2003	VinylSd	0.0	856.0	208500.0
1	1976	1976	MetalSd	0.0	1262.0	181500.0
2	2001	2002	VinylSd	0.0	920.0	223500.0
3	1915	1970	Wd Sdng	0.0	756.0	140000.0
4	2000	2000	VinylSd	0.0	1145.0	250000.0
5	1993	1995	VinylSd	0.0	796.0	143000.0
6	2004	2005	VinylSd	0.0	1686.0	307000.0
7	1973	1973	HdBoard	32.0	1107.0	200000.0
8	1931	1950	BrkFace	0.0	952.0	129900.0
9	1939	1950	MetalSd	0.0	991.0	118000.0

Task 2:

Data Exploration:

- Display the shape of the dataset
- Check data types of all columns
- Generate summary statistics of numerical features

Solution:

```
[ ] # 2-Data Exploration

# I. Display the shape of the dataset
print("\nShape of dataset:")
print(df.shape)

# II. Check data types of all columns
print("\nData Types:")
print(df.dtypes)

# III. Generate summary statistics of numerical features
print("\nSummary Statistics:")
print(df.describe())
```

Shape of dataset:
(2919, 13)

Data Types:

Id	int64
MSSubClass	int64
MSZoning	object
LotArea	int64
LotConfig	object
BldgType	object
OverallCond	int64
YearBuilt	int64
YearRemodAdd	int64
Exterior1st	object
BsmtFinSF2	float64
TotalBsmtSF	float64
SalePrice	float64
dtype:	object

Summary Statistics:

	Id	MSSubClass	LotArea	OverallCond	YearBuilt
count	2919.000000	2919.000000	2919.000000	2919.000000	2919.000000
mean	1459.000000	57.137718	10168.114080	5.564577	1971.312778
std	842.787043	42.517628	7886.996359	1.113131	30.291442
min	0.000000	20.000000	1300.000000	1.000000	1872.000000
25%	729.500000	20.000000	7478.000000	5.000000	1953.500000
50%	1459.000000	50.000000	9453.000000	5.000000	1973.000000
75%	2188.500000	70.000000	11570.000000	6.000000	2001.000000
max	2918.000000	190.000000	215245.000000	9.000000	2010.000000

Task 3:

Data Cleaning:

- Identify missing values
- Handle missing values appropriately
- Check and remove duplicate records if any

Solution:

```
[ ] print("\nMissing values:", df.isnull().sum())

# II. Handle missing values appropriately

# Drop records with null values
df = df.dropna() # you can also use fillna()

# Replacing SalePrice empty values with their mean values
df['SalePrice'] = df['SalePrice'].fillna(
    df['SalePrice'].mean())

# III. Check and remove duplicate records if any
df = df.drop_duplicates()

df.isnull().sum()
df.head()
```

Missing Values:

Id	0
MSSubClass	0
MSZoning	4
LotArea	0
LotConfig	0
BldgType	0
OverallCond	0
YearBuilt	0
YearRemodAdd	0
Exterior1st	1
BsmtFinSF2	1
TotalBsmtSF	1
SalePrice	1459
dtype:	int64

	Id	MSSubClass	MSZoning	LotArea	LotConfig	BldgType	OverallCond	YearBuilt	YearRemodAdd	Exterior1st	BsmtFinSF2	TotalBsmtSF	SalePrice
0	0	60	RL	8450	Inside	1Fam	5	2003	2003	VinylSd	0.0	858.0	208500.0
1	1	20	RL	9600	FR2	1Fam	8	1976	1976	MetalSd	0.0	1262.0	181500.0
2	2	60	RL	11250	Inside	1Fam	5	2001	2002	VinylSd	0.0	920.0	223500.0
3	3	70	RL	9550	Corner	1Fam	5	1915	1970	Wd Sdng	0.0	750.0	140000.0
4	4	60	RL	14280	FR2	1Fam	5	2000	2000	VinylSd	0.0	1145.0	250000.0

Task 4:

Feature Selection:

- Identify relevant features for analysis
- Remove unnecessary columns such as identifiers (e.g., Id)

Solution:

```
[ ] # 4-Feature Selection

# I. # Drop unnecessary columns (example: Id)
# Drop Id as it doesn't contribute in price
df.drop(['Id'],
        axis=1,
        inplace=True)
```

Task 5:

Data Preprocessing:

- Convert categorical variables into numerical form (encoding)
- Preparing the dataset for machine learning and display it's preview in the output.

Solution:

```
[ ] # 5-Data Preprocessing

# I. Convert categorical variables into numerical form (encoding)_(Covered in Supervised ML Part2)
cols = ['MSZoning', 'LotConfig', 'BldgType', 'Exterior1st']
df = pd.get_dummies(df, columns=cols, drop_first=True)

# II. Final dataset prepared for Machine Learning, preview:
print("\nProcessed Data:\n", df.head())
```

```
Processed Data:
  MSSubClass  LotArea  OverallCond  YearBuilt  YearRemodAdd  BsmtFinSF2  \
0         60    8450           5         2003          2003         0.0
1         20    9600           8         1976          1976         0.0
2         60   11250           5         2001          2002         0.0
3         70    9550           5         1915          1970         0.0
4         60   14260           5         2000          2000         0.0

  TotalBsmSF  SalePrice  MSZoning_FV  MSZoning_RH  ...  Exterior1st_CemntBd  \
0         856.0  208500.0        False        False  ...                False
1        1262.0  181500.0        False        False  ...                False
2         920.0  223500.0        False        False  ...                False
3         756.0  140000.0        False        False  ...                False
4        1145.0  250000.0        False        False  ...                False

  Exterior1st_MdBoard  Exterior1st_InStucc  Exterior1st_MetalSd  \
0                False                False                False
1                False                False                 True
2                False                False                False
3                False                False                False
4                False                False                False

  Exterior1st_Plywood  Exterior1st_Stone  Exterior1st_Stucco  \
0                False                False                False
1                False                False                False
2                False                False                False
3                False                False                False
4                False                False                False

  Exterior1st_VinylSd  Exterior1st_Md Sdng  Exterior1st_MdShing
0                 True                False                False
1                 False                False                False
2                 True                False                False
3                 False                 True                False
4                 True                 False                False
```